

Machine Learning-Assisted Digital Predistortion for Mask Margin Recovery in EML Transmitters

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Abstract We demonstrate that machine learning-assisted digital predistortion effectively recovers eye-mask margin in EML transmitters biased in nonlinear region. Experimental results at 25 GBd show a 13% margin improvement over the first order hold interpolation, while filter- and lookup-table-based approximations achieve comparable performance. ©2026 The Author(s) cengizhan.kaya@huawei.com

Introduction

The rapid expansion of short-reach optical links for data-center interconnects and access networks is driving the demand for higher data rates, lower power consumption, and simplified transmitter architectures. Electro-absorption modulated lasers (EMLs) are widely adopted in these applications due to their compact footprint, high modulation bandwidth, and suitability for photonic integration^{[1][2][3]}. Conventional EML-based transmitters, however, rely on an electrical amplifier (EA) to provide sufficient drive voltage for the electro-absorption modulator (EAM), ensuring an adequate extinction ratio (ER) and optical modulation amplitude (OMA). Since the EA contributes significantly to transmitter power consumption, cost, and complexity, driverless EML transmitter architectures could help reduce energy consumption.

In driverless operation, the reduced EAM drive voltage leads to a pronounced degradation in ER, which directly impacts receiver sensitivity. One approach to mitigate this is to lower the modulation bias voltage, thereby operating the EML in a more nonlinear regime to partially recover the ER. However, this strategy introduces significant nonlinear distortions, shifts the eye crossing point, and reduces the eye-mask margin. Since eye-mask compliance is a key requirement in standards such as ITU-T^[4] and IEEE 802.3^[5], recovering the ER alone is insufficient for standard-compliant operation.

To address these nonlinear effects, digital predistortion (DPD) has emerged as a promising solution. DPD techniques pre-compensate the transmitter waveform for EML nonlinearities and bandwidth limitations, thereby improving signal quality. Conventional DPD methods have demonstrated effectiveness in optical communication systems^{[6][7][8]}. However, most of these schemes optimize waveform-level metrics such as mean squared error (MSE) or error vector magnitude

(EVM), which may be suboptimal when eye compliance is the primary design objective.

In this paper, we propose a machine learning (ML)-DPD framework that directly optimizes the eye-mask margin without compromising the ER. A differentiable surrogate model of the transmitter chain, including the arbitrary waveform generator (AWG), EA, and EML, enables end-to-end optimization. A custom loss function explicitly linked to the eye-mask margin is introduced to mitigate nonlinear impairments. The proposed approach is demonstrated for a 25 GBd transmitter operating at 2 samples-per-symbol (sps). To enable practical implementation, the learned predistortion is approximated using a low-complexity finite impulse response (FIR) filter and a lookup table (LUT), preserving most of the performance gains while substantially reducing complexity.

Experimental Setup

Figure 1 depicts the experimental setup. The transmitter operates at a symbol rate of 25 GBd, employing PAM2 modulation with 2 sps data generation. A DPD module is used to optimize the transmitted signal. The digital signal is converted to an analog signal using an AWG running at 50 GSa/s. The signal is then amplified by an SHF S807 EA and fed to an EML (35 GHz bandwidth) emitting at 1310 nm. As the device is not designed for driverless operation due to the resulting low ER, an EA is employed after the AWG, while the operating point is intentionally shifted into the nonlinear region of the EML to highlight the effectiveness of DPD. The system is configured in an optical back-to-back setup using a 20 m single-mode fiber (SMF). At the receiver, a 50:50 optical splitter directs one output to an optical-to-electrical (O/E) converter and the other to a clock and data recovery (CDR) module for precise signal restoration. The outputs of these two modules are fed into a Keysight 86100D digital communication analyzer (DCA). Finally, the receiver's digital signal processing (DSP) block completes the process by performing DC-

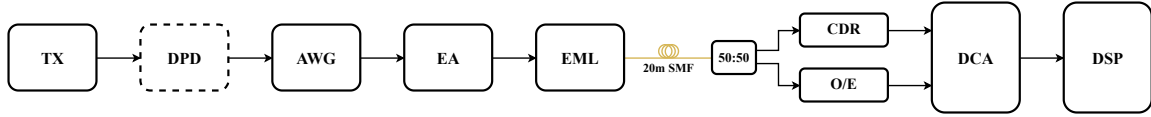


Fig. 1: The experimental setup. TX: transmitter; DPD: digital pre-distorter; AWG: arbitrary waveform generator; EA: electrical amplifier; EML: electro-absorption modulated laser; SMF: single-mode fiber; CDR: clock and data recovery; O/E: optical-to-electrical converter; DCA: digital communication analyzer; DSP: digital signal processing.

bias removal, power normalization of the PAM2 levels, and synchronization with the transmitted signal. It should be noted that metrics such as ER and eye-mask compliance are measured directly on the raw waveform using the DCA, prior to DSP processing; since the signal is only linearly amplified before digitization, these measurements remain unaffected, while the DSP is applied solely for subsequent modeling and analysis.

In the absence of DPD, upsampling from the symbol rate to 2 sps is realized by employing the first-order hold (FOH) interpolation technique^[9]. The algorithm sets the intermediate PAM2 sample to 0 during *symbol transitions* ($+1 \leftrightarrow -1$) and equal to the main sample for identical consecutive symbols. The driver swing and DC bias are tuned for a 10.7 dB ER and a -10% eye-mask margin to emulate a situation in which mask violation occurs due to modulator nonlinearity.

Sample-Based Digital Predistortion

To enable the application of machine learning-based optimization methods, the real-time experimental setup is approximated by a differentiable simulation model. A gradient-based optimization framework is then proposed, as illustrated in Fig. 2.

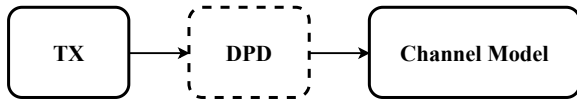


Fig. 2: The simulation setup. TX: transmitter; DPD: digital pre-distorter.

The implementation of this framework necessitates the ability to backpropagate gradients through all constituent models within the simulation, a requirement addressed through a three-stage process:

- **Surrogate Channel Model Training:** A differentiable channel model is first trained to accurately emulate the behavior of the physical channel.

The channel module is implemented as a learnable convolutional network that maps a 2 sps input waveform to a 16 sps output, enabling accurate EML link representation and mask margin loss evaluation. To ensure precise intra-symbol reconstruction, the architecture branches into 16 parallel convolutional heads, each predicting one sample of the 16 sps eye diagram, thus preserving temporal alignment and supporting symbol-synchronous learning. The resulting model achieves a normalized mean squared error (NMSE) of -20 dB.

- **Mask Margin Loss Implementation:** A custom *mask margin loss (MML)* function is developed to guarantee stable training while incorporating the desired margin standards.

A high-quality transmitter should produce a *wide-open eye* for better noise robustness and reliable bit decisions. This is defined by standardized forbidden (eye-mask) regions^[4]. To enforce compliance with the eye mask, we use a *Mask Margin Loss*, a differentiable function that penalizes signals entering the eye-mask region.

Let the model output be a one-dimensional vector of length L , $\mathbf{x} \in \mathbb{R}^L$. We partition $\mathbf{x}$ into S contiguous segments, each of length $P = 16$ for 16 sps channel output (so that $L = S \times P$). For each relative position p within a segment ($p \in \{1, \dots, P\}$), we define:

- a lower limit ℓ_p and upper limit u_p defining a prohibited interval (ℓ_p, u_p) ;
- a binary position mask $m_p \in \{0, 1\}$, where $m_p = 1$ indicates that the prohibited interval is active at position p , and $m_p = 0$ deactivates the penalty.

We use a logistic sigmoid with steepness parameter k to shape the loss: a larger k makes the sigmoid closer to a step function, while a smaller k yields smoother transitions and softer penalties.

For each segment s and position p within that segment, we introduce a continuous sample violation term $\nu_{s,p}$ that smoothly penalizes how deep the sample $x_{s,p}$ lies within the prohibited interval:

$$\nu_{s,p} = m_p \cdot \sigma(k(x_{s,p} - \ell_p)) \cdot \sigma(k(u_p - x_{s,p})). \quad (1)$$

The overall MML is the sum of all sample violation terms:

$$\mathcal{L}_{\text{MML}}(\mathbf{x}) = \sum_{s=1}^S \sum_{p=1}^P \nu_{s,p}. \quad (2)$$

- **DPD Model:** The sample-based digital pre-distortion (SB-DPD) is applied to the transmitted signal to perform the optimization.

The transmitter (TX) and DPD architecture are shown in Fig. 3. The transmitter output follows

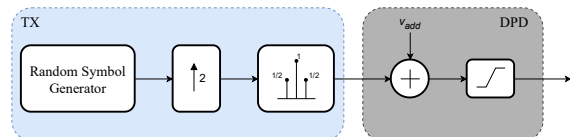


Fig. 3: SB-DPD module.

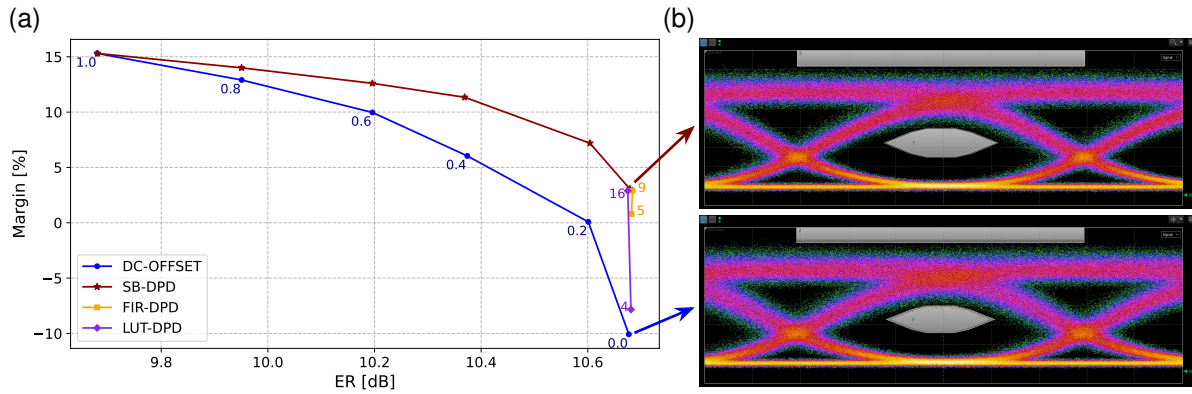


Fig. 4: (a) Experimental results showing the effect of DC-offset levels (blue), number of FIR taps (orange), and LUT sizes (purple). (b) DCA captures of eye diagrams.

the default configuration described in the experimental setup. The DPD generates an additive correction that is applied to the 2 sps waveform and clipped to $[-1, +1]$ to meet the AWG output limits. This clipping blocks gradient flow at the boundary values and, therefore, adapts only the transition samples, which are initially zero. The remaining samples remain fixed during the entire training process. Since the average value of the transmitted signal directly affects the ER and, consequently, the system operating point, the average value is constrained to remain constant throughout training to ensure stable optimization.

This architecture assumes full knowledge of the transmitted signal and directly determines the optimal pre-distorted values for eye-mask compliance. While impractical for real-time use, it serves as a reference to establish an upper bound on performance.

Practical DPD Structures

To approximate the behavior of the impractical SB-DPD system, three alternative methods were developed: a DC-offset method, an FIR-DPD, and a LUT-DPD method.

The DC-offset method for 2 sps PAM2 signals modifies the intermediate samples with a fixed DC offset when consecutive symbols differ ($+1 \leftrightarrow -1$).

The FIR-based method approximates the SB-DPD behavior using a linear filter with bias applied only to the intermediate samples. A large dataset of SB-DPD input-output pairs is collected, and the filter is trained in a least-squares sense to match the SB-DPD output, thereby approximating the nonlinear mapping with a linear convolutional model on the intermediate samples.

The LUT-based method uses a single long SB-DPD sequence to build a lookup table. For each pattern of local PAM2 symbols, the corresponding output values are averaged and stored. During operation, the predistorted output is selected based on the detected local symbol pattern, providing a data-driven, low-complexity approximation of the SB-DPD response.

Experimental Verification

For experimental verification, a PRBS-15 signal is used as the reference, and the results of 50 measurements are averaged for plotting.

In the DC-offset method, a fixed DC value is applied to the intermediate samples. Increasing the DC level enhances the eye-mask margin but reduces the ER, as the mean of the transmitted signal rises with larger DC offsets. As shown in Figure 4 (a), the improvement provided by SB-DPD over the DC-offset method decreases as the mean signal level increases; at a DC level of 1, SB-DPD and DC-offset outputs become identical, leaving no room for additional improvement. This demonstrates that DC-offset alone can partially linearize the signal, but SB-DPD provides better mask margin at each target ER, with diminishing returns as the signal mean increases.

For a DC level of 0, both FIR-DPD and LUT-DPD methods are additionally evaluated. FIR-DPD shows improved performance with increasing number of taps, and with 9 taps it closely matches SB-DPD performance. LUT-DPD with 4 entries (covering 2-symbol patterns) provides only modest improvement, whereas 16 entries (covering 4-symbol patterns) achieve performance comparable to SB-DPD. These results highlight the trade-off between complexity and linearization effectiveness for different DPD approximations.

Conclusion

SB-DPD effectively recovers eye-mask margin in nonlinear EML transmitters without degrading ER. Experimental results at 25 GBd and 2 sps show a 13% margin improvement over first-order hold interpolation. While simple DC-offset methods involve a trade-off between eye-mask margin and ER, SB-DPD further improves this by achieving higher eye-mask margins at each working point. Low-complexity FIR and LUT approximations closely replicate SB-DPD performance. These results demonstrate that practical approximations of SB-DPD enable high-performance pre-distortion for enhanced eye-mask compliance.

References

- [1] N. Kikuchi, R. Hirai, and T. Fukui, "Practical Implementation of 100-Gbit/s/Lambda Optical Short-Reach Transceiver with Nyquist PAM4 Signaling using Electroabsorptive Modulated Laser (EML)", in *2015 Optical Fiber Communication Conference*, 2015. DOI: 10.1364/OFC.2015.Th3A.2
- [2] T. Misawa, Y. Sugimoto, and K. Tanaka, "A Low-power, 128-Gbit/s, DC-coupled linear driver IC for Electroabsorption Modulated DFB Laser", in *2022 Optical Fiber Communication Conference*, 2022. DOI: 10.1364/OFC.2022.M2D.2
- [3] A. Uchiyama et al., "225 Gb/s PAM4 2 km and 10 km Transmission of Electro-Absorption Modulator Integrated Laser With Hybrid Waveguide Structure for 800 Gb/s and 1.6 Tb/s Transceivers", *Journal of Lightwave Technology*, vol. 42, no. 4, pp. 1225–1230, 2024. DOI: 10.1109/JLT.2023.3303884
- [4] "ITU-T Recommendation G.9804.3 (2021): 50-Gigabit-capable passive optical networks (50G-PON): Physical media dependent (PMD) layer specification", International Telecommunication Union - Telecommunication Standardization Sector (ITU-T), Recommendation G.9804.3, Sep. 2021. [Online]. Available: <https://www.itu.int/rec/T-REC-G.9804.3/en>
- [5] "IEEE Standard for Ethernet Amendment 9: Physical Layer Specifications and Management Parameters for 25 Gb/s and 50 Gb/s Passive Optical Networks", *IEEE Std 802.3ca-2020*, pp. 1–267, 2020. DOI: 10.1109/IEEESTD.2020.9135000
- [6] G. Khanna et al., "A Memory Polynomial Based Digital Pre-Distorter for High Power Transmitter Components", in *2017 Optical Fiber Communication Conference*, 2017. DOI: 10.1364/OFC.2017.M2C.4
- [7] G. Paryanti, H. Faig, L. Rokach, and D. Sadot, "A Direct Learning Approach for Neural Network Based Pre-Distortion for Coherent Nonlinear Optical Transmitter", *Journal of Lightwave Technology*, vol. 38, no. 15, pp. 3883–3896, 2020. DOI: 10.1109/JLT.2020.2983229
- [8] Z. Gong et al., "Transmitter Nonlinearity Mitigation Using Direct Learning Architecture Based Digital Predistortion Coefficients Identification", in *2024 Optical Fiber Communication Conference*, 2024. DOI: 10.1364/OFC.2024.Th3J.3
- [9] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 1996, pp. 768–774.